Evaluating a Risk-Managed Balanced Portfolio

An Investment Committee Analysis of Return, Downside Protection and Client Suitability

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Abstract

This project aims to evaluate whether an investment manager should offer a risk-managed version of a conventional 60/40 balanced portfolio. The proposed product retains the same strategic allocation of US equities and US government bonds, but applies a 10-month moving-average rule independently to each asset. When an asset finishes a month below its moving average, that allocation is transferred to a US Treasury bill proxy for the following month.

The analysis is framed around which investment strategy should be preferred for a wide range of investors rather than an exercise in selecting strategy with the highest back-tested returnt. The proposed strategy is assessed against monthly rebalanced 60/40 portfolio using return, volatility, drawdowns, recovery time, portfolio positioning, parameter sensitivity, turnover, transaction costs and monthly value attribution.

The 10-month overlay reduced volatility, from 9.61% to 6.94%, and reduced maximum drawdown from 20.51% to 14.94%. However, annual returns fell from 8.73% to 5.53% on average, the Sharpe ratio declined from 0.73 to 0.55, and it took longer to recover from the largest drawdown. The overlay was also substantially more expensive to implement as the turnover was approximately 215%, compared with 10% for the conventional portfolio.

The evidence therefore supports a cautious conclusion. The risk-management concept addresses a genuine client concern, but the 10-month specification does not currently provide a sufficiently strong risk-adjusted return. It could be useful to develop the strategy further as it seemed to add some value for the 40% bond allocation while the equity allocation tended to under-perform.

Investment Committee Recommendation

Recommendation: do not launch the 10-month moving-average portfolio in its current form as a replacement for the existing 60/40 balanced strategy.

The proposed overlay does achieve its intended objective in one important respect: it reduces the severity of historical losses and lowers portfolio volatility. However, the cost of this protection is substantial.

Over January 2015 to December 2025, the conventional 60/40 portfolio generated a CAGR of 8.73%, compared with 5.53% for the 10-month moving-average portfolio. A hypothetical $100,000 investment therefore grew to approximately $251,199 under the conventional strategy and $180,805 under the overlay before transaction costs.

The moving-average strategy reduced maximum month-end drawdown from -20.51% to -14.94%, but its Sharpe ratio was lower and its worst drawdown remained unrecovered for longer. At a base transaction-cost assumption of 5 basis points per dollar traded, the overlay's CAGR fell further to 5.30% and its final value to approximately $176,573.

The strategy should therefore not be presented as a superior balanced portfolio. Its potential use case is narrower: clients who are much more interested in reducing the magnitude of losses and are willing to sacrifice potential growth to achieve this.

The product concept may justify further research, but the current evidence is not strong enough to recommend that it is presented to clients.

1 Business Problem

A conventional balanced portfolio aims to combine the long-term growth potential of equities with the diversification and defensive characteristics of government bonds. A 60/40 stock–bond allocation is therefore a useful baseline for investors who want growth with less risk than an all-equity portfolio.

The weakness of a permanently invested allocation is that it continues to hold both assets during long periods of decline. This creates a potential client problem: some investors may have a sufficiently long amount of time to require meaningful growth, but insufficient tolerance for the magnitude or duration of losses that can occur in a fully invested portfolio.

The proposed product attempts to address this problem by introducing a systematic risk overlay. Rather than forecasting market direction, the rule reduces exposure after a sustained negative price trend has already developed.

2 Proposed Product

2.1 Conventional Balanced Portfolio

The baseline portfolio is allocated:

- 60% to US equities;
- 40% to US government bonds.

The portfolio is rebalanced monthly to restore the target allocation.

2.2 Risk-Managed Balanced Portfolio

The proposed portfolio begins from the same strategic 60/40 allocation.

At each month-end, SPY and IEF are compared independently with their respective 10-month moving averages. If an asset is above its moving average, its normal allocation is retained. If it is below its moving average, its allocation is transferred to BIL for the following month.

This creates four possible target positions:

Table 1: Possible target allocations under the proposed overlay

SPY	IEF	BIL	Portfolio State
60%	40%	0%	Fully invested
60%	0%	40%	Bond defensive
0%	40%	60%	Equity defensive
0%	0%	100%	Fully defensive

Signals are shifted forward by one month, so only information available at the relevant month-end is used to determine the following month's position. This prevents look-ahead bias.

3 Data and Backtest Design

The analysis uses adjusted closing prices for three US-listed exchange-traded funds:

Table 2: Asset proxies

Ticker	Role	Exposure
SPY	Equity allocation	S&P 500 / US large-cap equities
IEF	Bond allocation	7–10 year US Treasury bonds
BIL	Cash proxy	1–3 month US Treasury bills

Daily data are downloaded from January 2014 to December 2025 and converted to month-end values. The investment backtest begins in January 2015, giving 132 monthly observations. The earlier data are used only to establish the initial moving-average history.

Each strategy begins with a hypothetical $100,000 portfolio.

The S&P 500 is shown as a reference benchmark rather than as a proposed balanced-product alternative.

The full Python implementation is maintained separately at:

https://github.com/Louiskungfui/Investment-Strategy-Evaluation

3.1 Use of AI Assistance and Author Responsibility

AI tools were used extensively to support the development of the Python implementation, including code structuring, debugging and explanation of technical concepts. However, i selected and justified the modeling approach and verified the calculations. The interpretation, conclusions and final presentation are my own, and I retain sole responsibility for the accuracy of the paper.

4 Evaluation Criteria

The analysis focuses on measures that inform an investment-product decision.

Table 3: Decision metrics

Measure	Investment interpretation
CAGR	Long-term compound growth delivered to the client.
Annualised volatility	Overall variability of portfolio returns.
Maximum drawdown	Largest decline from a previous portfolio peak using month-end values.
Sharpe ratio	Excess return earned relative to variability of excess returns.
Calendar-year results	Provides an intuitive view of strong and weak annual outcomes.
Recovery time	Measures how long the investor remained below the previous portfolio peak.
Cash exposure	Shows how defensive the product is in practice rather than only in theory.
Turnover	Measures the implementation burden created by frequent portfolio changes.
Transaction-cost sensitivity	Tests whether the product remains economically attractive after realistic implementation friction.
Active-return attribution	Identifies which parts of the overlay created or destroyed value relative to the conventional portfolio.

5 Headline Performance

The figure below shows the growth of $100,000 under the conventional portfolio, the proposed 10-month overlay and the S&P 500 reference benchmark.

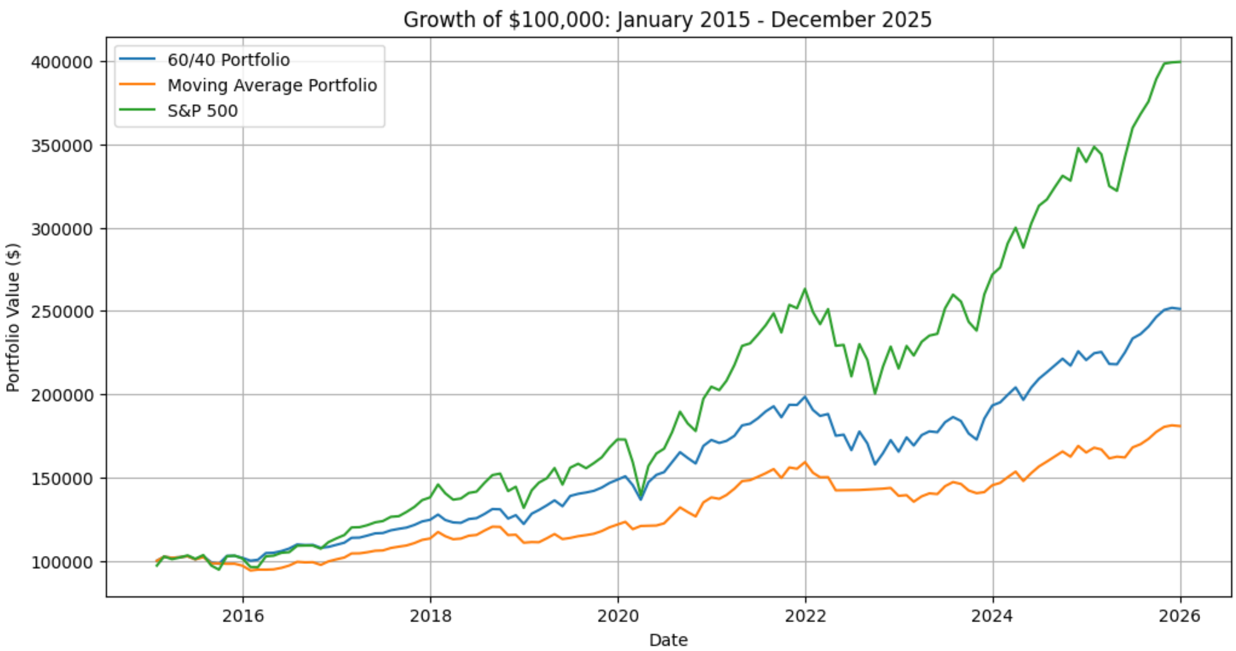


Table 4: Headline performance statistics

	Ann. Return	Ann. Vol.	Max DD	Sharpe	Final Value
60/40 Portfolio	8.73%	9.61%	-20.51%	0.73	$251,199
10-Month Overlay	5.53%	6.94%	-14.94%	0.55	$180,805
S&P 500	13.42%	14.96%	-23.93%	0.80	$399,529

The moving average strategy achieved a clear reduction in portfolio variability. Annualised volatility fell by 2.67 percentage points, while maximum month-end drawdown improved by 5.57 percentage points.

However, the return sacrifice was larger. CAGR fell by 3.20 percentage points and terminal wealth was approximately $70,393 lower than the conventional portfolio from the same $100,000 starting value.

The lower Sharpe ratio is particularly important. The strategy reduced risk, but the reduction in return was large enough that the overall risk-adjusted result was weaker under this measure.

6 Client Experience

Table 5: Client-focused performance statistics

	Best Year	Worst Year	Positive Months	Avg. Gain	Avg. Loss
60/40 Portfolio	21.82%	-16.65%	68.94%	2.18%	-2.46%
10-Month Overlay	15.36%	-12.72%	69.70%	1.48%	-1.85%
S&P 500	31.22%	-18.18%	69.70%	3.36%	-3.94%

The overlay produced a smaller worst calendar-year loss and a smaller average negative month. These are relevant advantages for a loss-sensitive client.

However, the average positive month also fell materially, from 2.18% to 1.48%. The defensive rule therefore changed both sides of the distribution: it reduced the typical loss, but also reduced participation in positive markets.

7 Drawdown Severity and Recovery

The figure below compares the month-end drawdown paths.

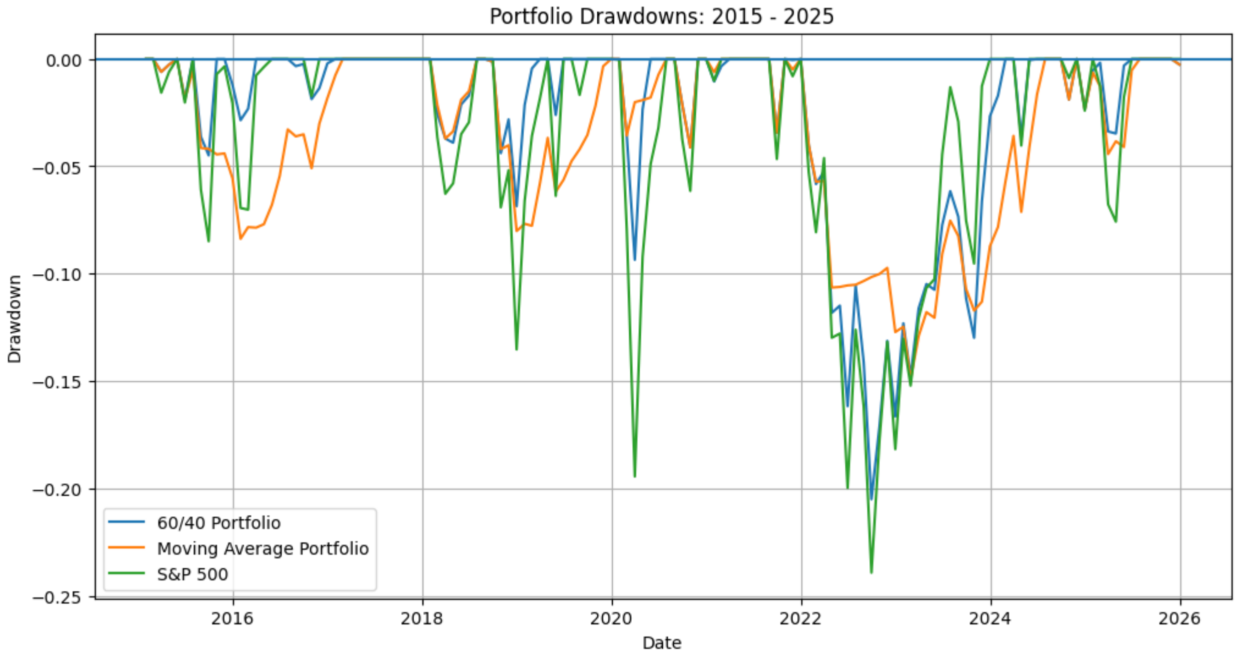


Table 6: Worst drawdown episode

	Peak	Trough	Max DD	Recovery	To Bottom	Total
60/40 Portfolio	Dec 2021	Sep 2022	-20.51%	Feb 2024	9 mo.	26 mo.
10-Month Overlay	Dec 2021	Feb 2023	-14.94%	Jul 2024	14 mo.	31 mo.
S&P 500	Dec 2021	Sep 2022	-23.93%	Dec 2023	9 mo.	24 mo.

The overlay reduced the severity of the worst drawdown, but it did not shorten the investor's time underwater. Its worst drawdown took 31 months from peak to recovery, compared with 26 months for the conventional 60/40 portfolio.

This is very important for product positioning. A client may prefer a shallower loss, but the risk-managed product cannot be described as delivering faster recovery on the evidence from this sample.

8 Parameter Robustness

The 10-month moving average was retained as the proposed specification as it was selected before the robustness analysis. Alternative periods were used to test whether the results were somewhat consistent to avoid basing analysis of an unusually weak/strong set of results.

Table 7: Moving-average sensitivity before transaction costs

Strategy	Ann. Return	Ann. Vol.	Max DD	Sharpe	Final Value
6-Month MA	6.35%	6.50%	-9.04%	0.70	$196,848
8-Month MA	6.45%	6.86%	-14.19%	0.68	$198,853
10-Month MA	5.53%	6.94%	-14.94%	0.55	$180,805
12-Month MA	6.60%	6.87%	-7.90%	0.70	$201,909
60/40 Portfolio	8.73%	9.61%	-20.51%	0.73	$251,199

The effect of the moving average strategy is consistent: every moving-average strategy reduced volatility and maximum drawdown relative to the conventional portfolio, while every specification also reduced CAGR.

The magnitude of the benefit is not robust however, the 12-month rule produced a substantially smaller maximum drawdown than the 10-month rule, with the 10-month rule produced the weakest return and Sharpe ratio of the four specifications.

This weakens the case for immediate product launch. Selecting the 12-month rule purely because it performed best does not guarantee better future performance.

9 How Defensive Is the Product?

The moving-average rule materially changes the portfolio's actual asset allocation.

Table 8: Portfolio positioning by moving-average length

	Fully Invested	Eq. Def.	Bond Def.	Fully Def.	Avg. Cash	Changes
6-Month	45.45%	9.85%	30.30%	14.39%	32.42%	50
8-Month	49.24%	8.33%	33.33%	9.09%	27.42%	42
10-Month	52.27%	8.33%	30.30%	9.09%	26.21%	44
12-Month	52.27%	9.09%	30.30%	8.33%	25.91%	35

Under the 10-month specification, the portfolio was fully invested in only 69 of 132 months. It was bond-defensive for 40 months, equity-defensive for 11 months and fully defensive for 12 months.

Average cash exposure was 26.21%.

This is important as the proposed product should not be communicated as a conventional balanced portfolio with a minor strategic adjustment. Over the sample, more than one quarter of the portfolio was held in the cash proxy on average.

A client purchasing the strategy would therefore need to understand that substantial periods of reduced market participation are an intentional feature of the product.

10 Turnover and Implementation Costs

A frictionless backtest can overstate the attractiveness of a strategy that changes allocations frequently.

The transaction-cost model therefore allows portfolio weights to drift with market returns during the month, calculates the trades required to reach the next target allocation, and applies a cost to the value bought and sold.

The base case assumes 5 basis points of cost per dollar traded (0.05%).

Table 9: Performance after a 5 bp transaction-cost assumption

	Net Return	Net Vol.	Max DD	Ann. Turnover	Total Costs	Final Value
60/40	8.72%	9.61%	-20.51%	9.95%	$168	$250,924
6-Month	6.09%	6.51%	-9.44%	246.87%	$3,423	$191,573
8-Month	6.24%	6.86%	-14.56%	193.51%	$2,935	$194,663
10-Month	5.30%	6.95%	-15.31%	215.27%	$2,990	$176,573
12-Month	6.41%	6.87%	-8.04%	172.22%	$2,444	$198,119

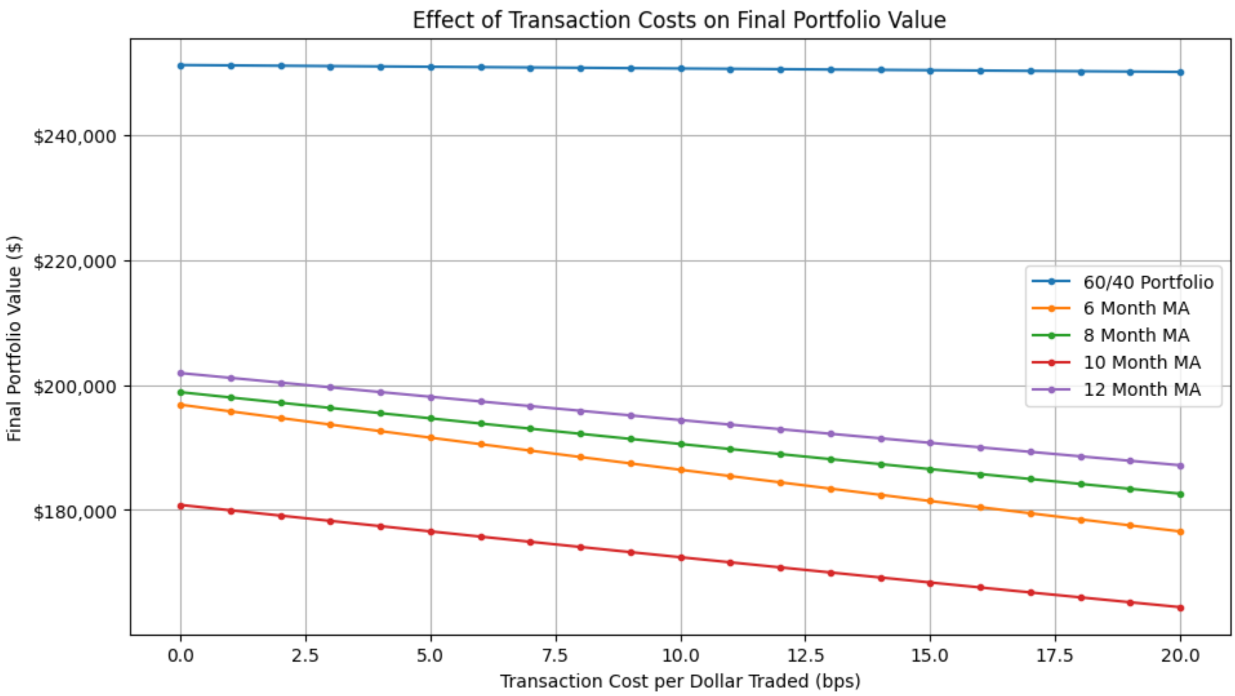
The implementation burden is therefore different for each strategy.

The conventional portfolio generated annualised one-way turnover of only 9.95%. The 10-month overlay generated approximately 215%. At the 5 bp base cost, the resulting cumulative transaction costs were approximately $2,990 for the overlay compared with only $168 for the conventional portfolio.

Transaction costs do not explain the entire performance gap, but they increase it.

11 Transaction-Cost Sensitivity

The base transaction-cost assumption should not be treated as precise. Sensitivity analysis is therefore performed from 0 to 20 basis points.



The relationship between assumed cost and terminal wealth is approximately linear over the range tested.

Table 10: Terminal-value sensitivity to each additional basis point of trading cost

Strategy	Loss per Additional 1 bp	Estimated Loss at +10 bp
6-Month MA	\$1,014	\$10,141
10-Month MA	\$818	\$8,178
8-Month MA	\$812	\$8,123
12-Month MA	\$737	\$7,375
60/40 Portfolio	\$55	\$549

The linear fits all produced an R^2 rounded to 1.00 in the 0–20 bp sensitivity range.

For the headline 10-month strategy, each additional basis point of assumed trading cost reduced terminal wealth by approximately \$818 across the full backtest. The corresponding sensitivity for the conventional portfolio was only approximately \$55.

This provides a direct commercial interpretation of turnover: the overlay is more exposed to implementation assumptions.

12 Where the Overlay Added and Destroyed Value

The active return of the overlay is defined as its monthly return minus the return of the conventional 60/40 portfolio.

The difference can be decomposed into:

- the effect of replacing the 60% equity allocation with BIL when equities are defensive; and
- the effect of replacing the 40% bond allocation with BIL when bonds are defensive.

Table 11: Active return by 10-month portfolio state

State	Months	Avg. Active	Sum Active	Avg. Overlay	Avg. 60/40
Bond Defensive	40	0.03%	1.30%	0.51%	0.48%
Equity Defensive	11	-2.31%	-25.42%	0.30%	2.61%
Fully Defensive	12	-0.95%	-11.37%	0.16%	1.11%
Fully Invested	69	0.00%	0.00%	0.53%	0.53%

The most important finding is the difference between the equity and bond components.

Across the monthly attribution, the equity overlay contributed an arithmetic sum of -36.26 percentage points of active return, while the bond overlay contributed +0.78 percentage points. The total arithmetic sum of active monthly returns was -35.48 percentage points.

Because returns compound, these arithmetic sums should not be interpreted as a terminal return difference. On a compounded basis, the 10-month overlay finished approximately 28.02% below the conventional portfolio.

The evidence suggests that the equity de-risking rule was the main source of underperformance in this sample, while the bond rule was approximately neutral to modestly beneficial.

This does not prove that an equity overlay is structurally unattractive. It does, however, provide a clear direction for further product research: the equity and bond components should be evaluated separately rather than assuming that the same trend rule is equally effective for both asset classes.

13 Stress Protection Versus Rebound Risk

The monthly attribution illustrates the central trade-off of the product.

Table 12: Selected months where the overlay added or destroyed value

Month	State	60/40	Overlay	Active Return
Sep 2022	Fully Defensive	-7.44%	0.21%	+7.65%
Mar 2020	Equity Defensive	-6.00%	1.61%	+7.61%
Jun 2022	Fully Defensive	-5.29%	0.08%	+5.37%
Aug 2022	Fully Defensive	-3.99%	0.19%	+4.18%
Apr 2020	Equity Defensive	7.73%	0.10%	-7.63%
Nov 2023	Fully Defensive	7.30%	0.47%	-6.83%
Jul 2022	Fully Defensive	6.71%	0.04%	-6.67%
Oct 2015	Equity Defensive	4.85%	-0.27%	-5.12%

The strategy can provide substantial protection when a negative trend continues after the defensive signal has been triggered. September 2022 and March 2020 are the strongest examples in the sample.

The same mechanism creates significant rebound risk. In April 2020, the conventional portfolio gained 7.73% while the overlay gained only 0.10%, producing -7.63 percentage points of active return in one month.

The product therefore does not eliminate market-timing risk. It changes its form. The client receives protection against some sustained declines but accepts the risk of remaining defensive during rapid recoveries.

This is one of the most important points for client communication.

14 Client Suitability

The proposed strategy should not be marketed on the basis that it is a superior version of 60/40. Its historical return was lower, its Sharpe ratio was lower and implementation was more expensive.

Its potential target client is more specific.

14.1 Potential Target Client

A plausible target investor would:

- require exposure to long-term growth assets;
- place unusually high value on limiting the depth of portfolio losses;
- be willing to hold substantial cash allocations for extended periods;
- accept underperformance during rapid market rebounds;
- accept lower expected long-term growth in exchange for a smoother return path; and
- understand that a lower drawdown does not guarantee a faster recovery.

14.2 Clients for Whom the Product Is Less Suitable

The strategy appears less appropriate for investors whose primary objective is maximising long-run capital growth and who have sufficient capacity to remain invested through large temporary losses.

It may also be unsuitable for a client who expects a "balanced portfolio" to remain predominantly invested in equities and bonds. The 10-month strategy held approximately 26% in BIL on average, so its defensive behaviour is a central product feature rather than an occasional exception.

15 Product and Governance Considerations

Several issues would need to be addressed before any launch decision.

15.1 Parameter Governance

The 10-month specification should not be replaced by the 12-month rule simply because the latter produced a better historical result. Doing so after observing the backtest would create a data-mining concern.

Any revised parameter should be justified economically and tested using a process that separates development data from validation data.

15.2 Implementation Capacity

The overlay's turnover is much higher than that of the conventional portfolio. Actual implementation would therefore require more detailed modelling of bid–ask spreads, market impact, execution timing and fund size.

15.3 Client Communication

Clients would need to understand that defensive positioning can cause substantial relative underperformance during recoveries. A risk-managed label should not imply either capital protection or guaranteed downside limits.

15.4 Product Differentiation

The attribution suggests that the bond overlay and equity overlay did not contribute equally. A future product-development process should therefore test whether a simpler or more targeted rule can retain downside protection with lower opportunity cost and turnover.

16 Limitations

The analysis is intentionally practical, but several limitations remain.

- The backtest covers a single eleven-year investment period.
- Maximum drawdown is measured using month-end portfolio values and may understate intra-month losses.
- SPY, IEF and BIL are ETF proxies rather than complete representations of all equities, government bonds and cash instruments.
- Transaction costs are modelled as a constant number of basis points per dollar traded. Actual costs vary through time, by security and by order size.

- Taxes, management fees and market impact are not included.
- The moving-average sensitivity analysis uses the same historical sample as the headline test and is therefore not a substitute for true out-of-sample validation.
- The analysis is nominal and does not adjust returns for inflation.
- Historical drawdowns and volatility do not impose limits on future losses.
- Client profiles are simplified and do not replace a full suitability assessment including objectives, liabilities, liquidity needs, time horizon and capacity for loss.

17 Further Work Before Launch

If the Investment Committee wished to continue development, the next stage should focus on validation rather than adding more backtest complexity.

Priority work would include:

1. perform a longer-history or walk-forward validation of the proposed rules;
2. measure daily rather than month-end drawdowns while retaining monthly trading signals;
3. evaluate the equity and bond overlays separately;
4. test actual implementation assumptions using representative bid–ask spreads and portfolio sizes;
5. assess whether reduced turnover can preserve most of the downside benefit;
6. evaluate the strategy against explicit client objectives, such as a maximum tolerated drawdown or required minimum return.

18 Conclusion

The moving-average overlay demonstrates a genuine investment principle: systematic de-risking can reduce the severity of some portfolio losses, but downside protection is not free.

The 10-month strategy reduced annualised volatility from 9.61% to 6.94% and improved maximum month-end drawdown from -20.51% to -14.94%. It also reduced the average negative month and protected capital effectively during several of the most adverse months in the sample.

However, these benefits came with significant costs. CAGR fell from 8.73% to 5.53%, the Sharpe ratio declined, recovery from the worst drawdown took longer, and the strategy generated approximately 215% annualised turnover. At a 5 bp transaction-cost assumption, terminal wealth fell further to approximately $176,573.

The attribution analysis provides the clearest explanation. The bond defensive rule was approximately neutral to modestly positive, whereas equity de-risking was the principal source of historical underperformance, particularly when the strategy remained defensive during sharp recoveries.

The Investment Committee should therefore not approve the 10-month overlay as a replacement for the conventional 60/40 product on the evidence currently available.

The stronger conclusion is more nuanced: the product concept addresses a real client need, but its current design imposes too large a return and implementation cost for the level of downside protection achieved. The appropriate next step is targeted product development and out-of-sample validation rather than launch.

A Calendar-Year Returns

Table 13: Calendar-year returns, 2015–2025

Year	60/40 Portfolio	10-Month Overlay	S&P 500
2015	1.67%	-3.04%	1.23%
2016	7.73%	3.96%	12.00%
2017	13.71%	12.42%	21.71%
2018	-2.03%	-2.25%	-4.57%
2019	21.82%	9.83%	31.22%
2020	16.04%	13.42%	18.33%
2021	15.06%	15.36%	28.73%
2022	-16.65%	-12.72%	-18.18%
2023	16.78%	4.58%	26.18%
2024	14.12%	13.53%	24.89%
2025	13.95%	9.61%	17.72%

B Transaction-Cost Sensitivity

Table 14: Selected transaction-cost sensitivity results

Strategy	Cost	Ann. Return	Ann. Vol.	Max DD	Total Cost	Final Value
60/40	0 bp	8.73%	9.61%	-20.51%	$0	$251,199
60/40	5 bp	8.72%	9.61%	-20.51%	$168	$250,924
60/40	10 bp	8.71%	9.61%	-20.52%	$336	$250,649
10-Month	0 bp	5.53%	6.94%	-14.94%	$0	$180,805
10-Month	5 bp	5.30%	6.95%	-15.31%	$2,990	$176,573
10-Month	10 bp	5.08%	6.96%	-15.68%	$5,905	$172,437

C Interpretation of Attribution Sums

The overlay-contribution figures reported in the main paper are arithmetic sums of monthly active-return contributions. They are useful for identifying which component of the rule created or destroyed value, but they should not be interpreted as compounded terminal-return differences.

The compounded relative performance measure is calculated separately. Over the full backtest, the 10-month overlay finished approximately 28.02% below the conventional 60/40 portfolio on a compounded relative basis.